

Phase 3: Application Selection for Performance Testing (Week 3):

1. Select applications representing different workload types (e.g. CPU-intensive, RAM intensive, I/O-intensive, Network-intensive, and Server applications such as game servers) for performance evaluation and create an Application Selection Matrix listing applications with justifications for choosing them.?

Ans: The select application representing different workload types are follows:

Workload type	Selected Application	Justification
CPU-intensive	Stress-ng	used to assess CPU capacity and for plans as maintaining a heavy CPU use.
RAM intensive	Memory test	offers controlled memory allocation for testing memory availability and management.
i/O-intensive	fio	I create disc read/write operations to examine disc I/O and archive memory speed.
Network-intensive	lperf3	evaluates the timing and speeds of the network.
Sever application	nginx	Correct server workloads are simulated by a lightweight web server.

These applications were picked because they are common, simple, and work well for testing in a virtual Ubuntu Server setup.

1. Installation Documentation with exact commands for SSH-based installation?

Ans:

1.Sudo apt update:

```
ramesh@rameshserver:~$ sudo apt update
[sudo] password for ramesh:
Hit:1 http://gb.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://gb.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://gb.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

The computer's software listing was revised using a command like `Sudo apt update`. The output shows that the server's packages are current, showing that it is ready for app installs, and that the Linux sources was visited.

`Sudo apt install stress-ng -y:`

Installing the stress-ng package took place via the Ubuntu package manager. The output confirms which the present version is the most modern one and it says the archive arrives and working.

`Sudo apt install fio -y:`

```
Preparing to unpack .../10-librdmacm1t64_50.0-2ubuntu0.2_amd64.deb ...
Unpacking librdmacm1t64:amd64 (50.0-2ubuntu0.2) ...
Selecting previously unselected package librados2.
Preparing to unpack .../11-librados2_19.2.3-0ubuntu0.24.04.1_amd64.deb ...
Unpacking librados2 (19.2.3-0ubuntu0.24.04.1) ...
Selecting previously unselected package libpmemobj1:amd64.
Preparing to unpack .../12-libpmemobj1_1.13.1-1.1ubuntu2_amd64.deb ...
Unpacking libpmemobj1:amd64 (1.13.1-1.1ubuntu2) ...
Selecting previously unselected package librbd1.
Preparing to unpack .../13-librbd1_19.2.3-0ubuntu0.24.04.1_amd64.deb ...
Unpacking librbd1 (19.2.3-0ubuntu0.24.04.1) ...
Selecting previously unselected package fio.
Preparing to unpack .../14-fio_3.36-1ubuntu0.1_amd64.deb ...
Unpacking fio (3.36-1ubuntu0.1) ...
Setting up libboost-thread1.83.0:amd64 (1.83.0-2.1ubuntu3.1) ...
Setting up libnbd0 (1.20.0-1) ...
Setting up libglusterfs0:amd64 (11.1-4ubuntu0.1) ...
Setting up libboost-iostreams1.83.0:amd64 (1.83.0-2.1ubuntu3.1) ...
Setting up libdaxctl1:amd64 (77-2ubuntu2) ...
Setting up libndctl6:amd64 (77-2ubuntu2) ...
Setting up librdmacm1t64:amd64 (50.0-2ubuntu0.2) ...
Setting up libpmem1:amd64 (1.13.1-1.1ubuntu2) ...
Setting up libgfxdr0:amd64 (11.1-4ubuntu0.1) ...
Setting up librados2 (19.2.3-0ubuntu0.24.04.1) ...
Setting up libpmemobj1:amd64 (1.13.1-1.1ubuntu2) ...
Setting up librbd1 (19.2.3-0ubuntu0.24.04.1) ...
Setting up libgfrpc0:amd64 (11.1-4ubuntu0.1) ...
Setting up libgfapi0:amd64 (11.1-4ubuntu0.1) ...
Setting up fio (3.36-1ubuntu0.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
```

Sudo apt install iperf3 -y:

```
Selecting previously unselected package libiperf0:amd64.  
(Reading database ... 125638 files and directories currently installed.)  
Preparing to unpack .../libiperf0_3.16-1build2_amd64.deb ...  
Unpacking libiperf0:amd64 (3.16-1build2) ...  
Selecting previously unselected package iperf3.  
Preparing to unpack .../iperf3_3.16-1build2_amd64.deb ...  
Unpacking iperf3 (3.16-1build2) ...  
Setting up libiperf0:amd64 (3.16-1build2) ...  
Setting up iperf3 (3.16-1build2) ...  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

Using the Ubuntu package managers, Iperf3 was installed. The output shows that all required interactions were properly opened and setup, and no computer services needed to be resumed after the install.

Sudo apt install nginx -y:

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ramesh@rameshserver:~$ sudo apt install iperf3 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iperf3 is already the newest version (3.16-1build2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ramesh@rameshserver:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 564 kB of archives.
After this operation, 1,596 kB of additional disk space will be used.
Get:1 http://gb.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://gb.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 1s (644 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 125654 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ramesh@rameshserver:~$
```

2. Expected Resource Profiles documenting anticipated resource usage?

Ans: Estimated Use of Resources

stress (CPU tests):

This is meant to improve CPU usage on one or more cores. Testing planning and the system's ability to manage high loads is the aim.

stressful (tests of memory):

To test memory planning, caching, and if memory is available since needed, this requires significant amounts of RAM.

Fio:

This is meant to put a lot of reads and writes on the disc. It will check the file system's operation.

Iperf3:

This should boost the volume of data passing in the network, enabling tests of delay, speed, and data packet handling.

Expected to consume moderate CPU and memory resources while processing the requests for simulating actual server scenarios.

These profiles create a minimum standard for future comparison.

4. Monitoring Strategy explaining measurement approach for each application?

Ans: System performance can be remotely viewed using SSH command-line tools. The monitoring approach that will be followed is:

CPU Usage: Used to analyse the system's scheduling activities and processing workload during CPU-intensive tasks.

Memory Usage: Monitored to gauge memory pressure and availability under RAM-intensive workloads.

Disk Activity: Analysed during disk I/O benchmarking for reading or writing performance.

Network Performance: Tested on the network to measure its bandwidth and stability.

Overall System Load: Tested in the server application environment for performance measurement.

Measurements will be taken before, during, and after the execution of the workload to enable comparison and analysis.